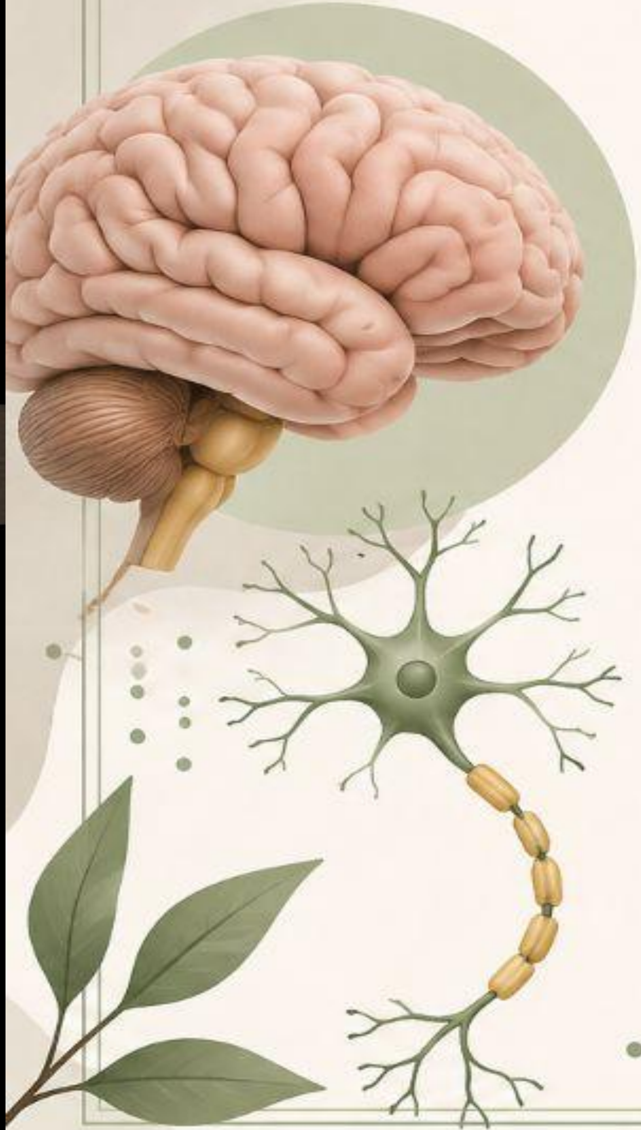
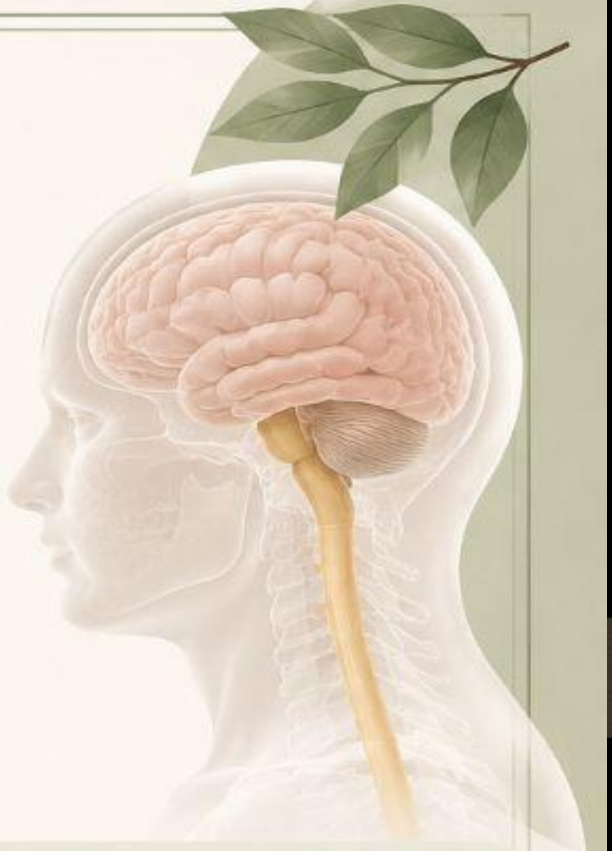


Introduction to CNS

Central Nervous System

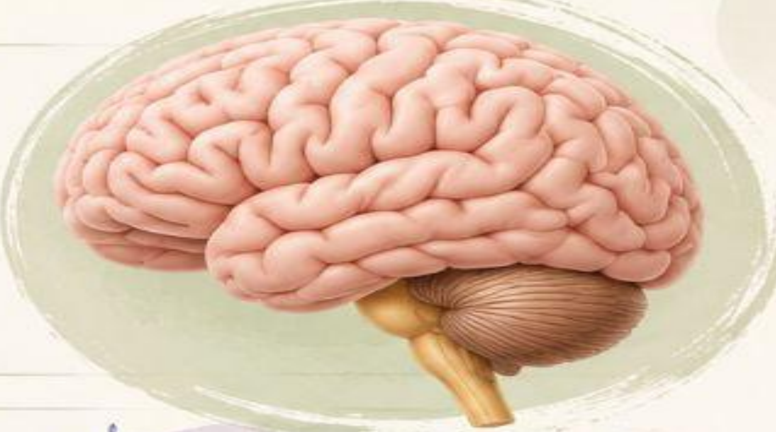
Understanding the control center of our body and mind



- Controls thoughts, emotions and behavior
- Processes and transmits information
- Coordinates body functions and responses
- Maintains balance and homeostasis

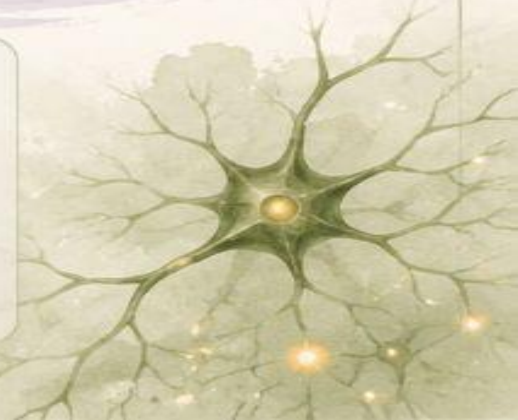
THE NERVOUS SYSTEM

- The nervous system is the **master controlling and communicating system** of the body.



- The nervous system **controls and coordinates** all essential functions of human body.

Controls thoughts, emotions and behavior	Processes and transmits information	Coordinates body functions and responses	Maintains balance and homeostasis



FUNCTION OF NERVOUS SYSTEM

The nervous system has **3 main functions:**

1. Sensory Function (Input)

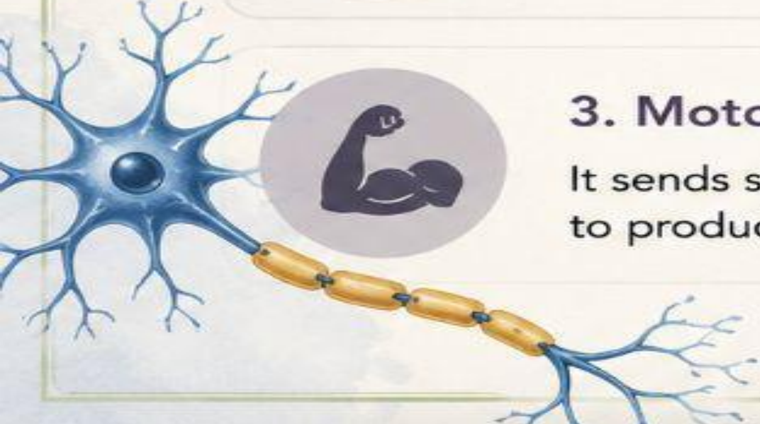
It receives information from the internal and external environment.

2. Integrative Function (Processing)

It processes, interprets and stores the information.

3. Motor Function (Output)

It sends signals to muscles or glands to produce a response.



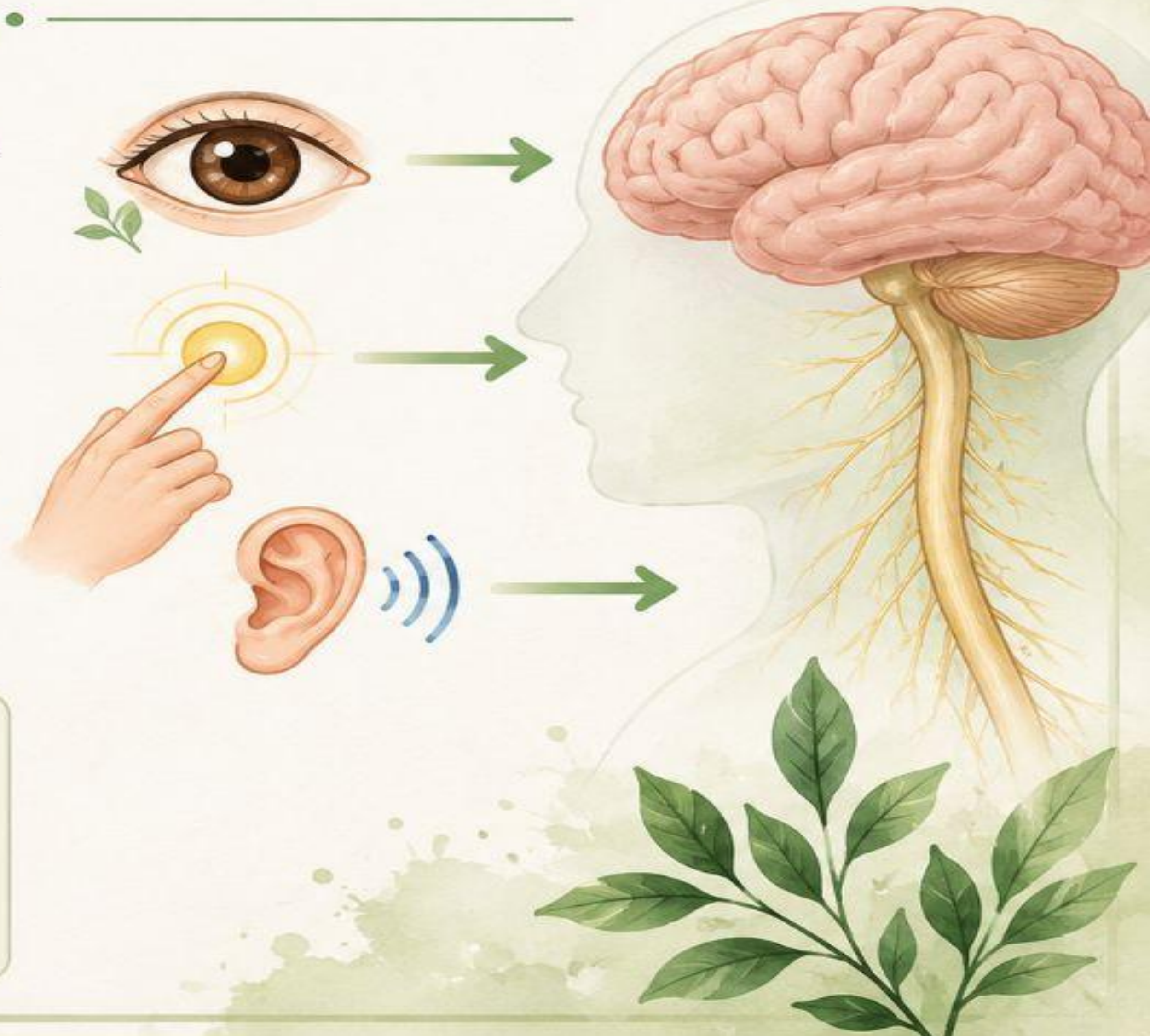
SENSORY INPUT

The first step of nervous system function

These changes are called **stimuli**.

A **stimulus** is any change inside or outside the body that can be detected by **sensory receptors**.

Gathered information is called **sensory input**.



Easy Definition

The information collected by sensory receptors and sent to the brain is called **sensory input**.

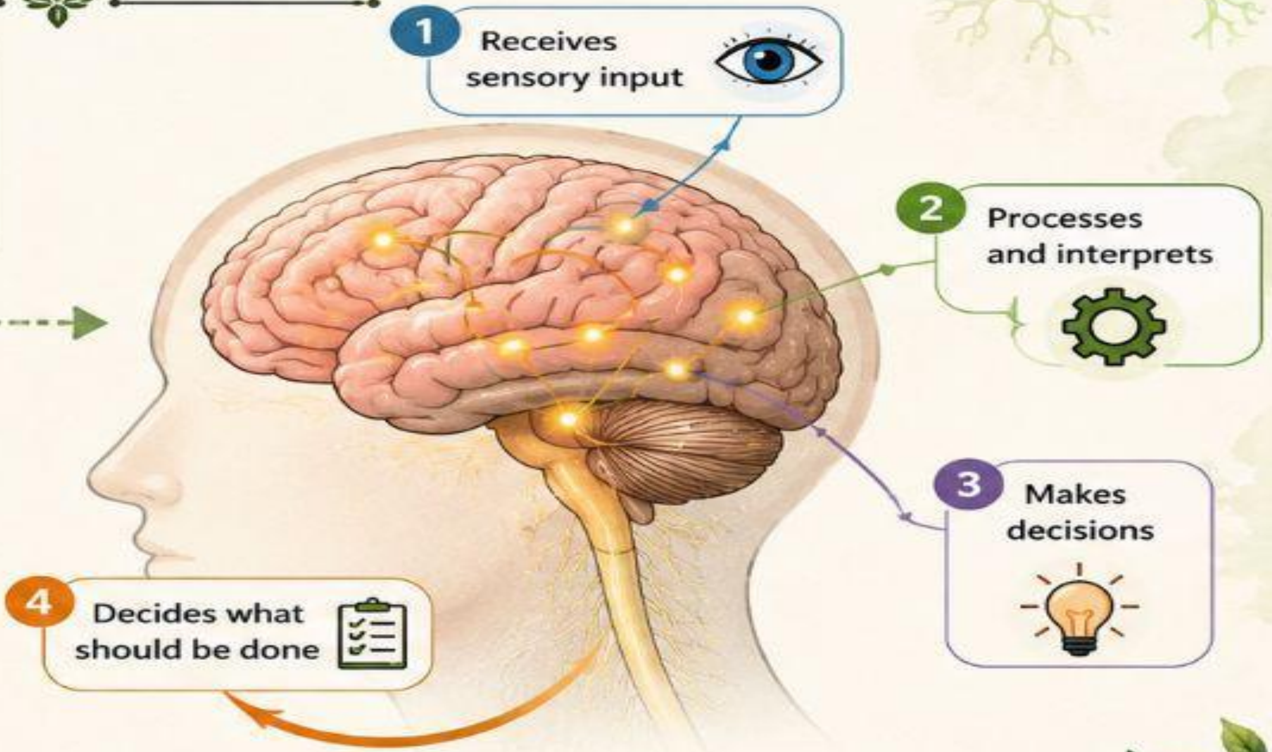
INTEGRATIVE FUNCTION
(PROCESSING)

The nervous system **processes** and **interprets** the sensory input and **makes decisions** about what should be done.

After receiving sensory input, the **brain thinks.**

It asks:

- What happened?
- Is it dangerous?
- What should I do?
- This is called the **integrative function.**



What happened?	Is it dangerous?	What should I do?
Understanding the situation	Assessing the risk	Choosing the right action

This is called the **integrative function.**

Real-Life Example

Example Situation:
You touch a hot pan.

Step 1 – Sensory Function

Skin detects heat. → Signal goes to the brain. → Information reaches the brain.

Sensory receptors in the skin send signals through nerves to the brain.

Step 2 – Integrative Function

Brain understands: "The pan is hot." → Brain decides: "Remove your hand immediately."

The brain **processes** the information, thinks about it, and decides what to do.

Step 3 – Motor Function (Output)

Signal sends from brain to muscles. → Muscles receive the signal. → You pull your hand away quickly.

The muscles act and produce a response to **protect** you.

This whole process happens in seconds to keep you safe.

What happens next?

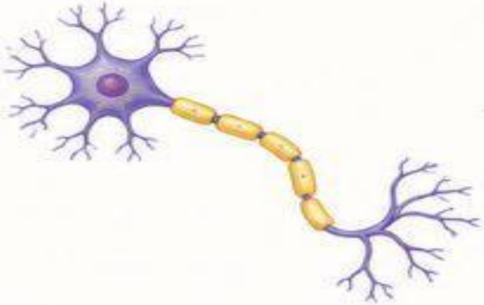
The brain sends a command to the muscles.

This is called the **Motor Function**.

1. Command is sent

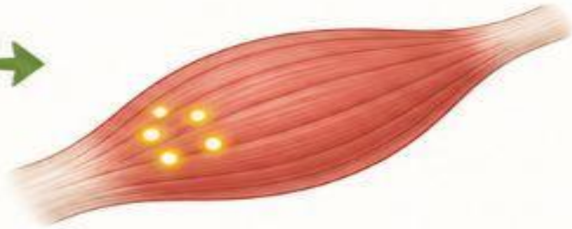
The brain sends an electrical command to the muscles through the spinal cord.

2. Signal travels



The signal travels along nerves to reach the muscles.

3. Muscles receive the signal



The muscles receive the signal and activate.

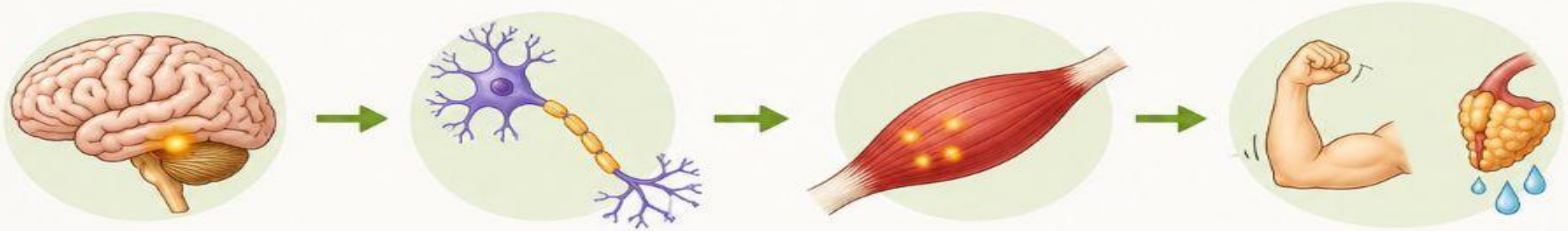
4. Action occurs

The muscles contract and produce movement or an action.

✓ This process is very fast and happens in a fraction of a second to help you respond and act.

FUNCTION OF NERVOUS SYSTEM

- **MOTOR FUNCTION:** The nervous system send signals to muscle, glands and organs so they can respond correctly such as muscular contraction and glandular secretion.



1. Signal sent
The brain sends an electrical command to the muscles, glands, or organs.

2. Signal travels
The signal travels along nerves to reach the target.

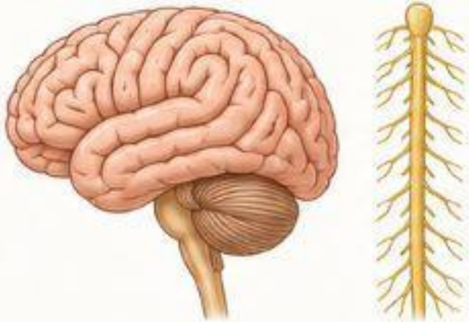
3. Target responds
The muscles contract or the glands/organs become active.

4. Action occurs
The body produces the correct response (movement or secretion).

These signals allow the body to respond quickly and properly to keep us safe and functioning.

STRUCTURAL CLASSIFICATION OF NERVOUS SYSTEM

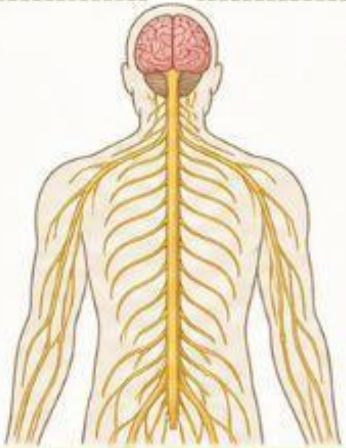
- **CENTRAL NERVOUS SYSTEM** ----brain and spinal cord which act as command centers of nervous system.



COMMAND CENTERS

They receive, process, and send information.

- **PERIPHERAL NERVOUS SYSTEM:** Part of nervous system outside CNS. They link all parts of body by carrying impulses from sensory receptors to CNS and from CNS to appropriate gland or muscle.



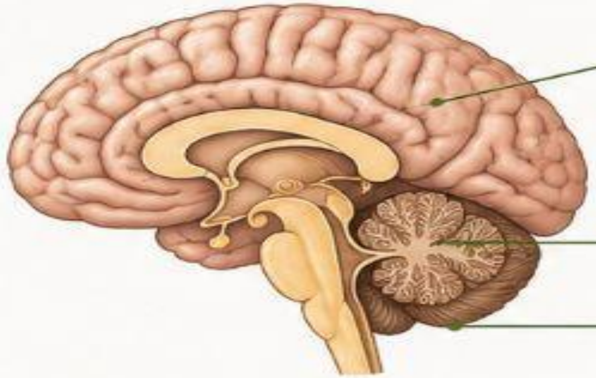
COMMUNICATION NETWORK

They connect the CNS to all parts of the body and help coordinate responses.

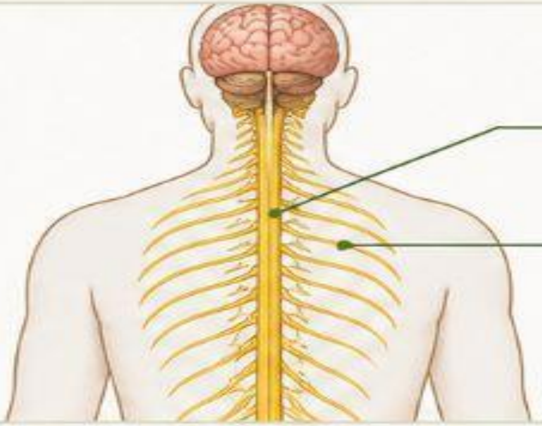
Together, these two parts allow the body to sense, think, and respond.

CENTRAL NERVOUS SYSTEM

- **BRAIN:**
  - Located inside cranial cavity.
  - Consists of Cerebral Hemisphere, brainstem and Cerebellum.



- **SPINAL CORD:**
  - Spinal cord is a reflex center and conduction pathway.



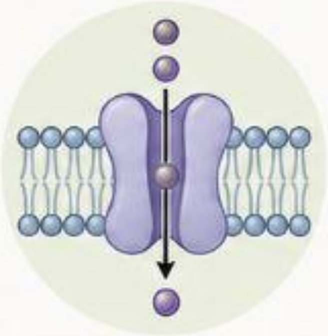
The brain and spinal cord work together to receive, process, and send signals throughout the body.

TARGETS OF CNS DRUG ACTION

What does "Target" mean?

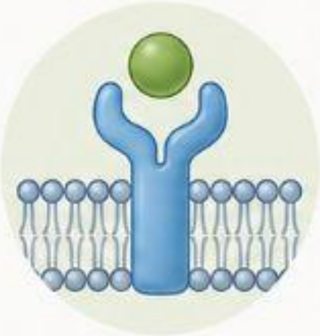
A **target** is the place where a **drug acts** to produce its effect.

Examples of targets:



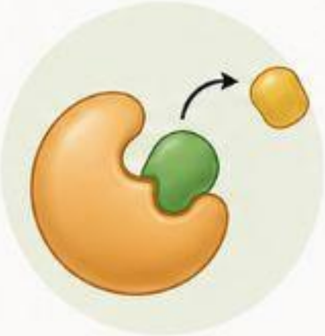
Ion channels

Proteins that control the flow of ions across cell membranes.



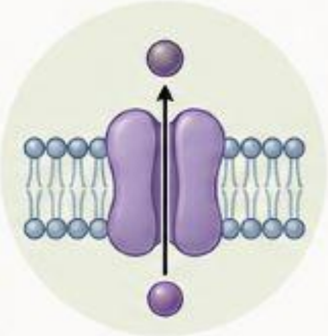
Receptors

Proteins that bind signaling molecules and start a response inside the cell.



Enzymes

Proteins that speed up chemical reactions in the cell.



Neurotransmitter transporters

Proteins that remove neurotransmitters from the synaptic cleft.

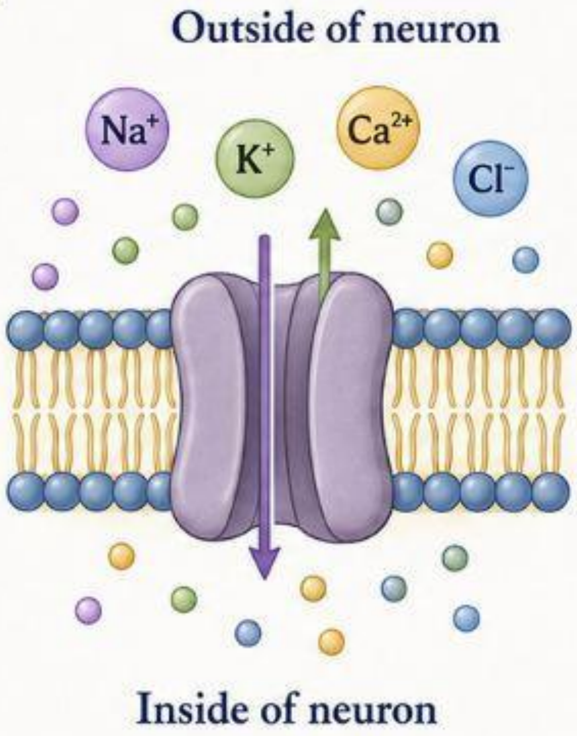
Most drugs that act on the central nervous system (CNS) appear to do so by changing ion flow through transmembrane channels of nerve cells.

Ions are **electrically charged particles**.

The most important ions in the nervous system are: **Na, K, Ca, Cl**

These ions move **into** and **out of neurons**.

This movement creates **electrical signals (nerve impulses)**.



By affecting how these ions move, drugs can influence the activity of neurons and change how the brain and nervous system function.

Ion flow means:

➡ **Movement** of **ions** across the **cell membrane.**

Example

When sodium enters the neuron,

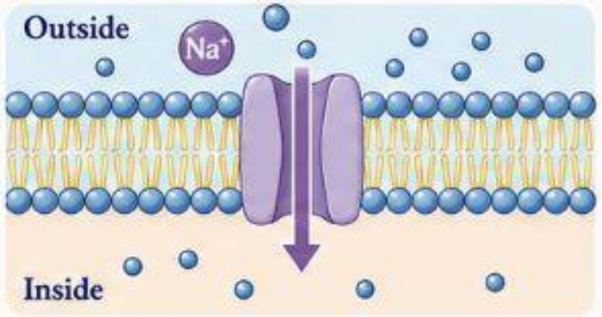
↓

The neuron becomes excited.

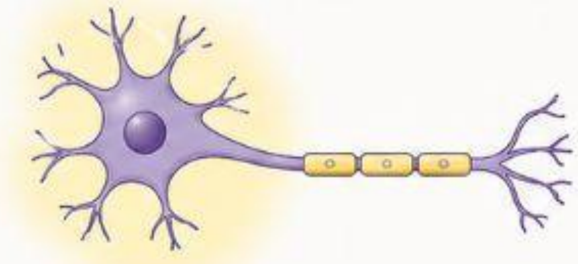
↓

A nerve impulse is produced.

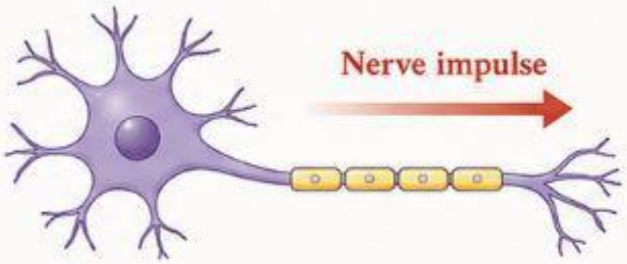
1. Sodium ions (Na^+) move from outside into the neuron through ion channels.



2. The neuron becomes excited (depolarized).

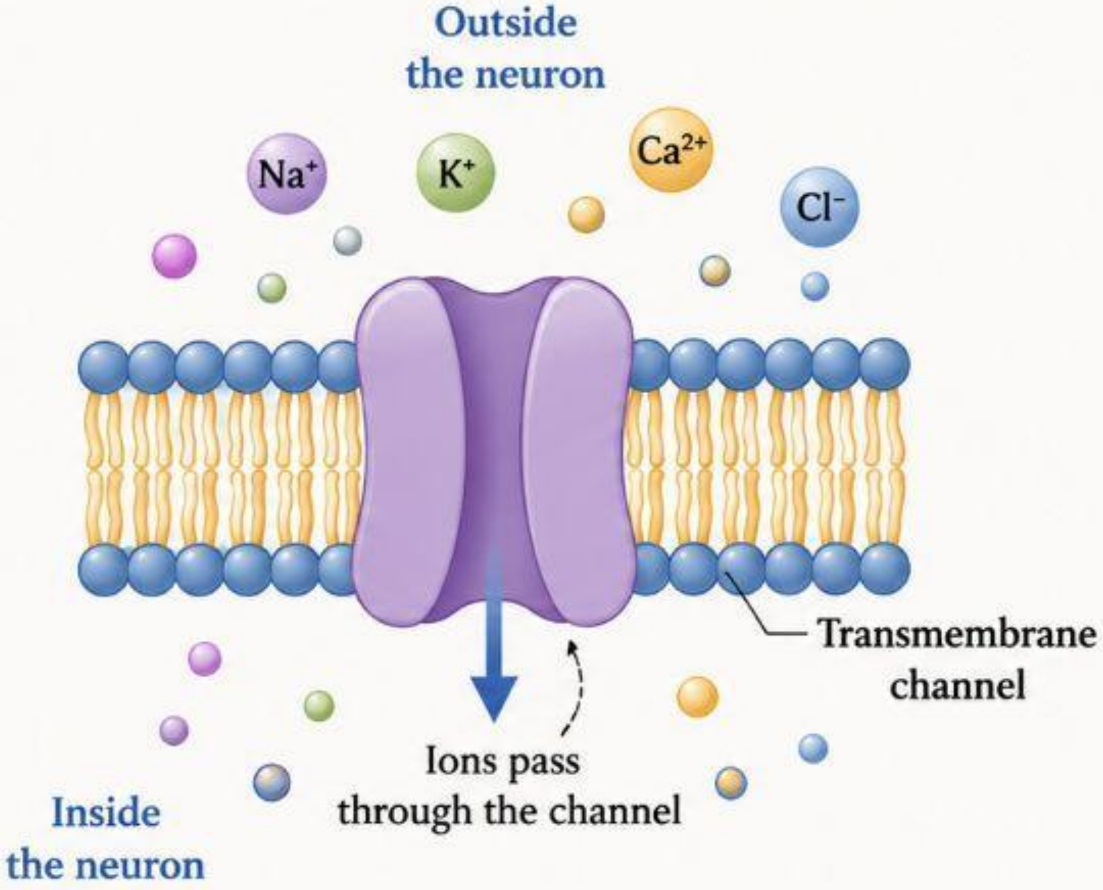


3. A nerve impulse is produced.



Transmembrane channel

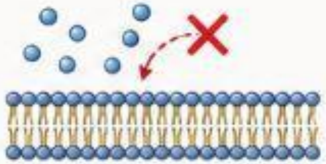
- A **channel** that passes through the cell membrane.
- Think of it as a **door** in the neuron's membrane.
- Only certain **ions** can pass through this door.



Why are ion channels important?

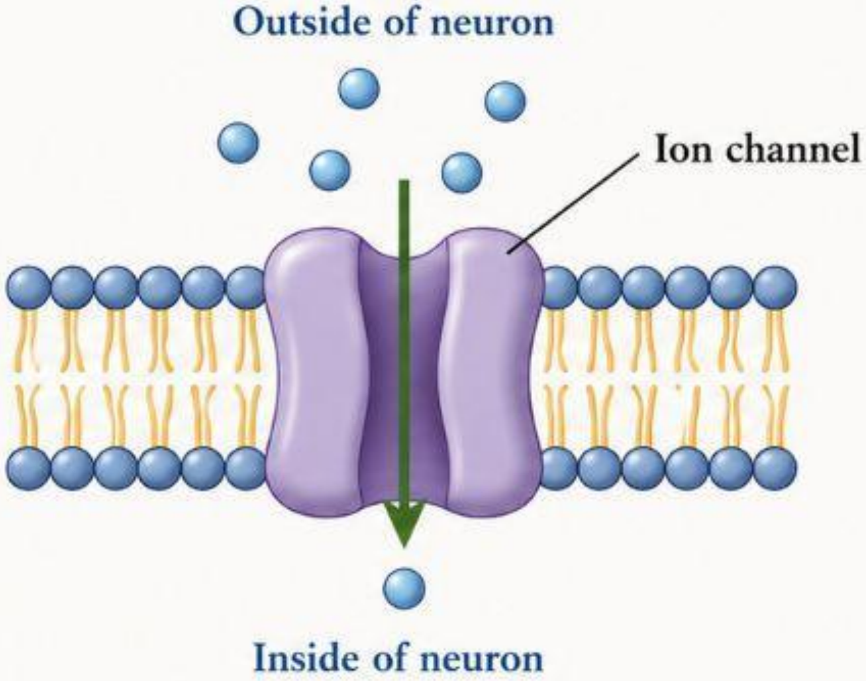
Without ion channels:

- No **sodium** entry



- No **nerve impulse**

- No **communication** between neurons



So ion channels are **essential** for normal nervous system function.

Types of Ion Channels

Ion channels in neurons are of **2 main types**

1. Voltage-Gated Ion Channel

Trigger: Change in **electrical charge (voltage)**

Opens when the membrane **voltage** changes.

Example: Sodium (Na^+) channels open during an action potential.

Easy Memory Tip:
Voltage = Electricity

2. Ligand-Gated Ion Channel

Trigger: **Chemical** messenger **(Ligand)**

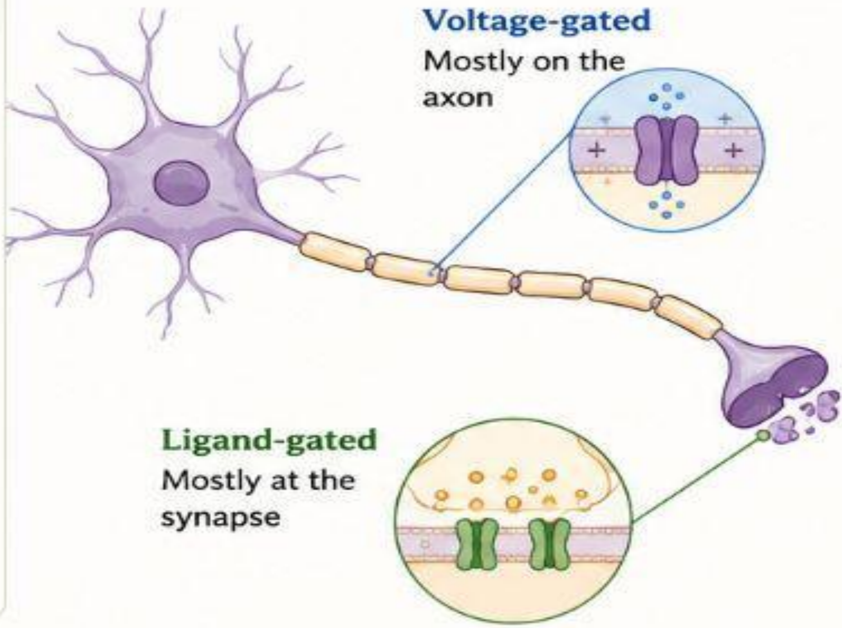
Opens when a **neurotransmitter** binds to the receptor.

Examples of ligands:

- Acetylcholine (ACh)
- GABA
- Glutamate

Easy Memory Tip:
Ligand = Chemical Key

Where are they found?



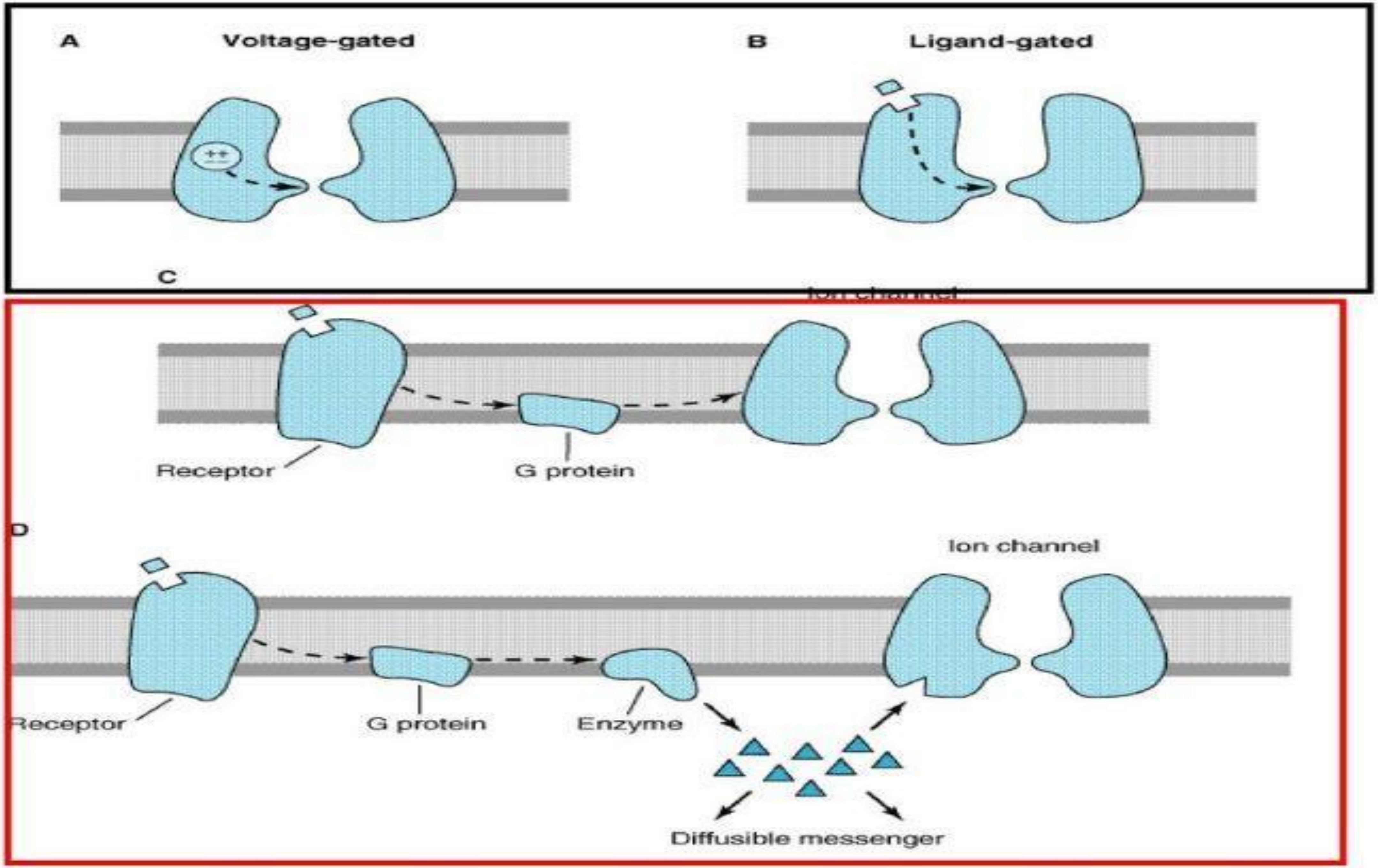
Quick Comparison

	Voltage-Gated	Ligand-Gated
Opens by	Electrical change	Neurotransmitter (chemical)
Trigger	Voltage	Ligand
Location	Mainly on axon	Mainly at synapse
Function	Produces action potential	Passes signals between neurons

Remember This!

Voltage = Electricity → Opens voltage-gated channels

Ligand = Chemical → Opens ligand-gated channels


A
Voltage-gated
B
Ligand-gated
C
Receptor
G protein
D
Ion channel
Receptor
G protein
Enzyme
Diffusible messenger

Voltage-gated ion channels

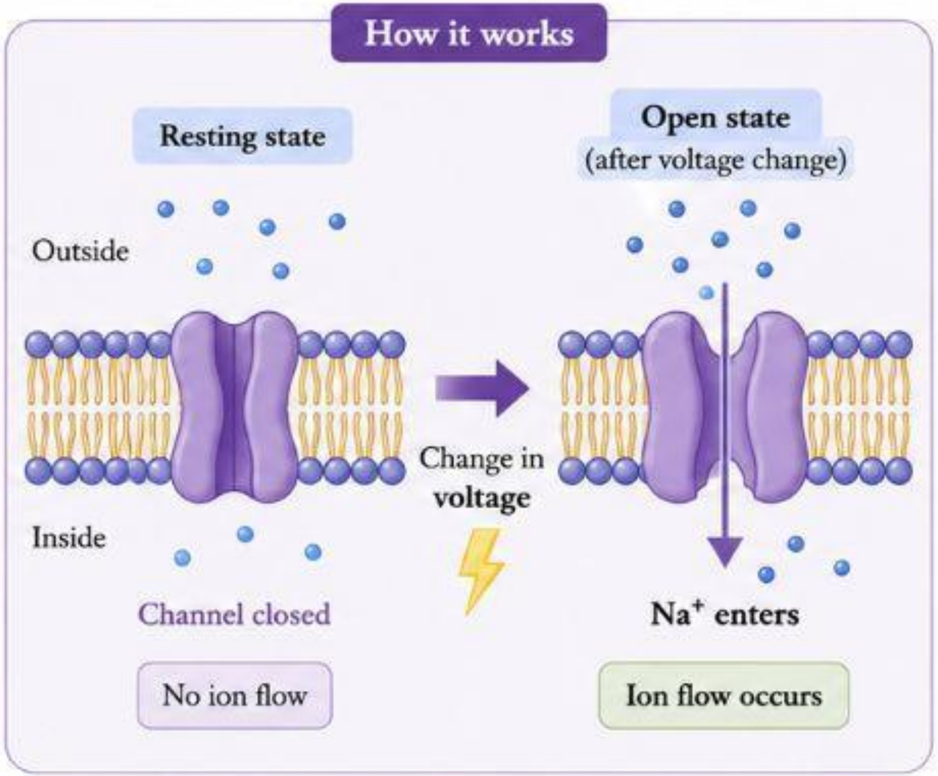
- Voltage-gated ion channels are **doors** in the nerve cell membrane that open when the **electrical charge** (**voltage**) changes.

- They are concentrated on the **axons** of nerve cells and include the **sodium channels** responsible for action potential propagation.

Example:

Change in voltage → Sodium channels open → Na^+ enters → Action potential propagates

Key Point: *These channels allow ions to move across the membrane in response to a* ***change in membrane potential.***



Remember: *Voltage* change = *Channel* ***opens*** = ***Ions move***

1. What is Voltage?

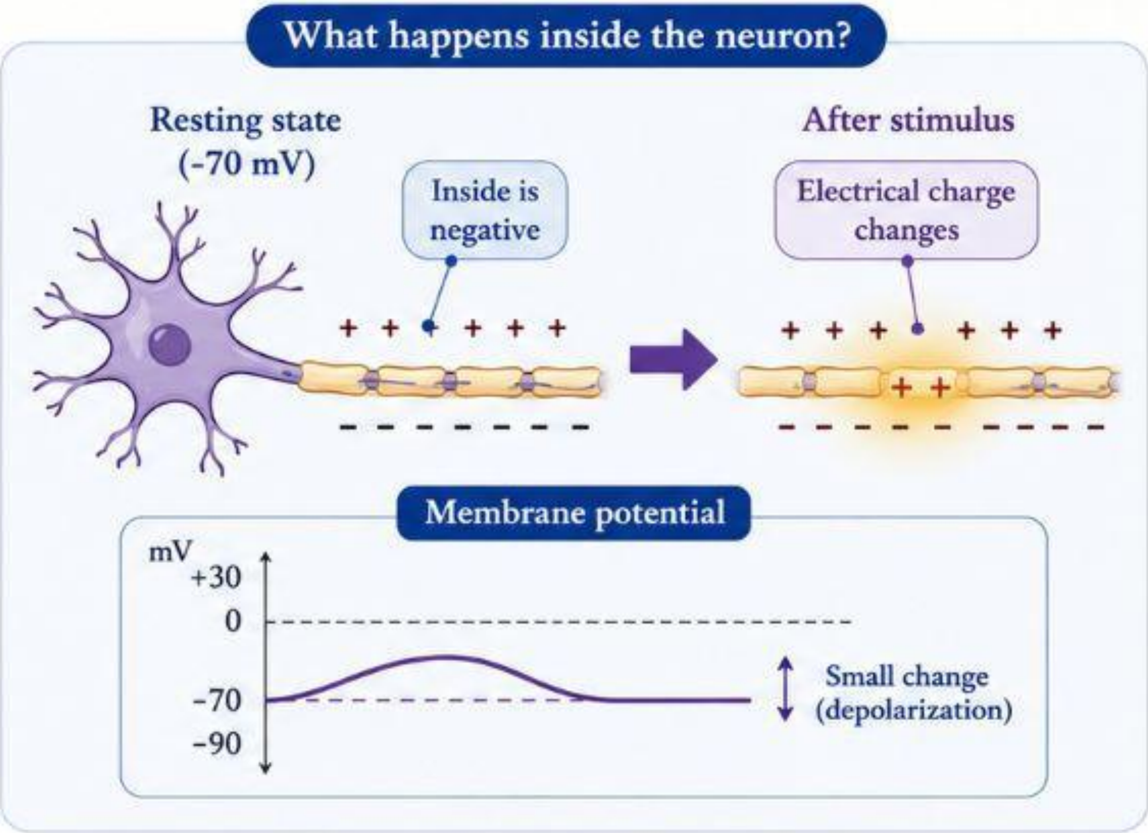
Voltage means **electrical charge**.

Every neuron has a small **electrical charge**.

Normally, the neuron is at **resting membrane potential** (about **-70 mV**).

When a stimulus comes, the **electrical charge changes**.

This change is called **voltage change**.



Key Point: Voltage is the electrical change across the neuron's membrane. It is the basis of all nerve signals.

Remember: Resting potential is about **-70 mV**.

What is Membrane Potential?

Every **neuron** has:

Positive charges (+) outside.

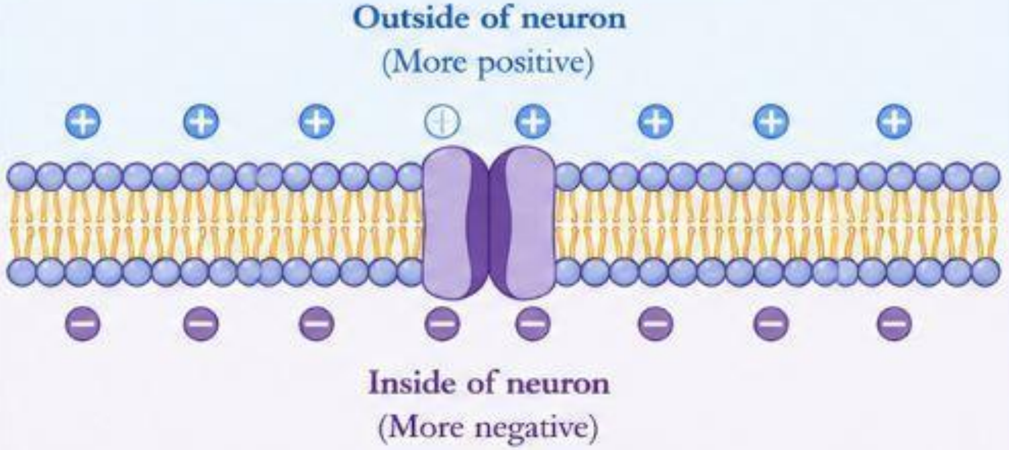
More negative charges (−) inside.

This difference in charge is called the **membrane potential.**

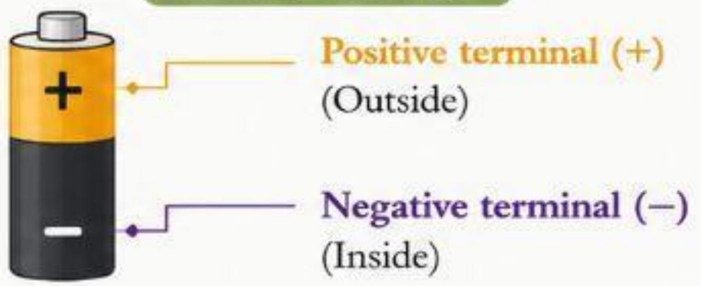
Think of it like a battery

- A battery has a positive (+) and a negative (−) terminal.
- Similarly, a neuron also has electrical charges.

Inside and outside the neuron



Battery Analogy

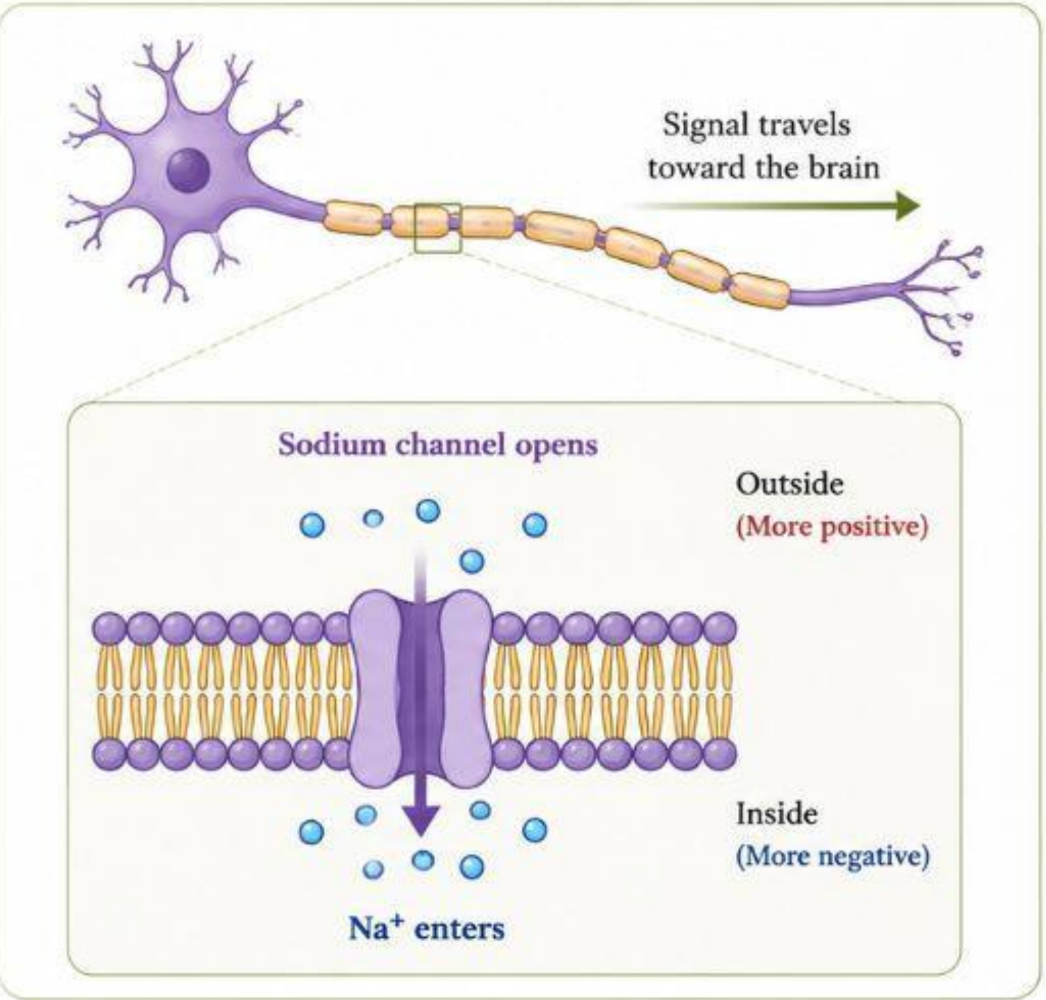


Key Point: Membrane potential is the electrical difference across the neuron's membrane.

It is the basis for all **nerve activity.**

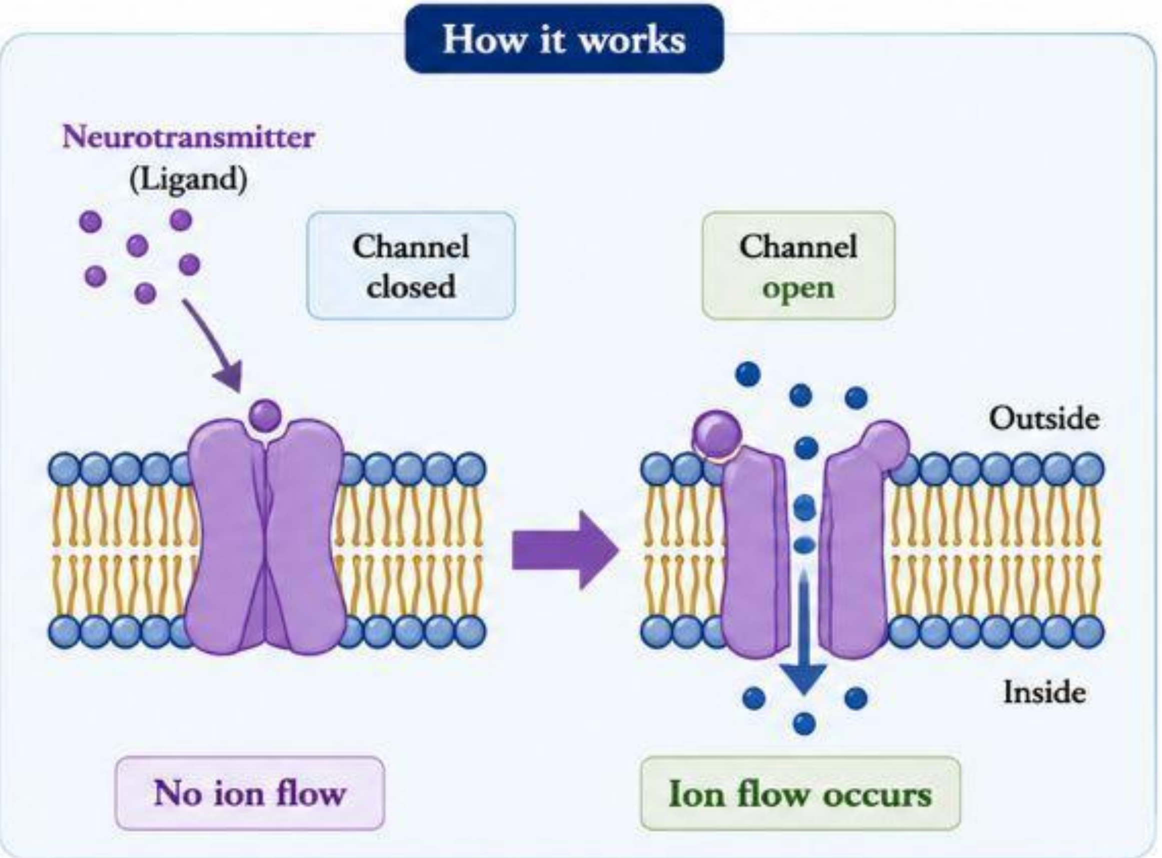
Example

1. You touch a **hot cup**.
2. Heat **stimulates** the neuron.
3. **Electrical charge** changes.
4. Voltage-gated **sodium channels** open.
5. **Sodium** enters.
6. The inside of the neuron becomes less negative (**depolarization**).
7. The **nerve impulse** starts.



Ligand-gated ion channels

- **Ligand-gated ion channels** are channels that open when a **neurotransmitter** (**chemical messenger**) binds to them.



The neurotransmitter (ligand) acts like a **key**.

When it binds to the receptor, the channel **opens**.

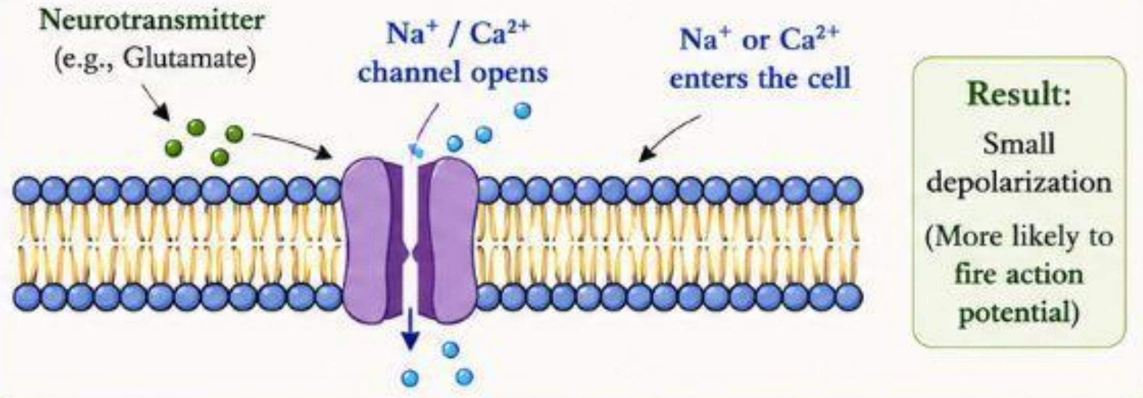
Ions flow through the channel and the **signal is passed** to the next neuron.

Key Point: Ligand-gated channels open in response to a chemical messenger (neurotransmitter). They help transmit signals between neurons.

Role of the Ion Current Carried by the Channel

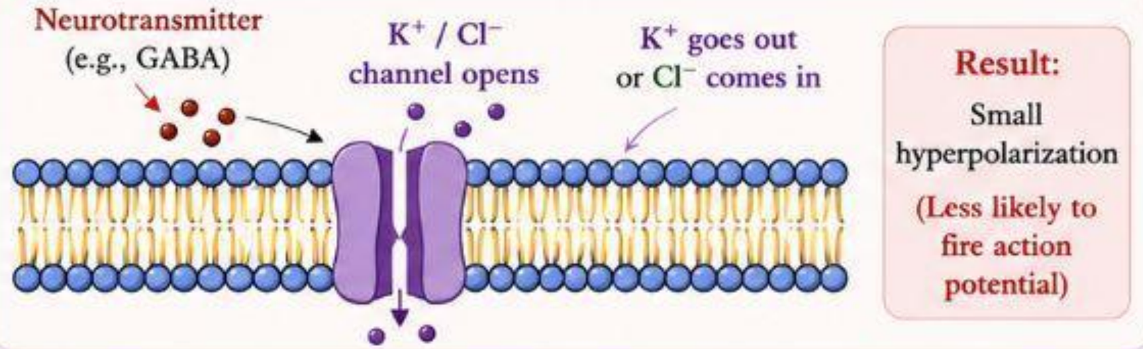
Excitatory postsynaptic potentials (EPSPs)

- Are usually generated by the opening of **sodium (Na^+)** or **calcium (Ca^{2+})** channels.
- An **EPSP** is a small electrical change that makes the neuron **more likely to fire an action potential.**



Inhibitory postsynaptic potentials (IPSPs)

- Are usually generated by the opening of **potassium (K^+)** or **chloride (Cl^-)** channels.



Key Point: EPSPs make the neuron more likely to fire, while IPSPs make it less likely to fire. The **balance** between EPSPs and IPSPs determines whether the neuron will fire an action potential or not.

How is an EPSP produced?

1. A **neurotransmitter** (e.g., glutamate) binds to its **receptor**.

↓

2. The **sodium (Na^+)** or **calcium (Ca^{2+})** channel opens.

↓

3. Na^+ or Ca^{2+} enters the neuron.

↓

4. The inside of the neuron becomes less negative (**depolarization**).

↓

5. The neuron becomes **excited**.

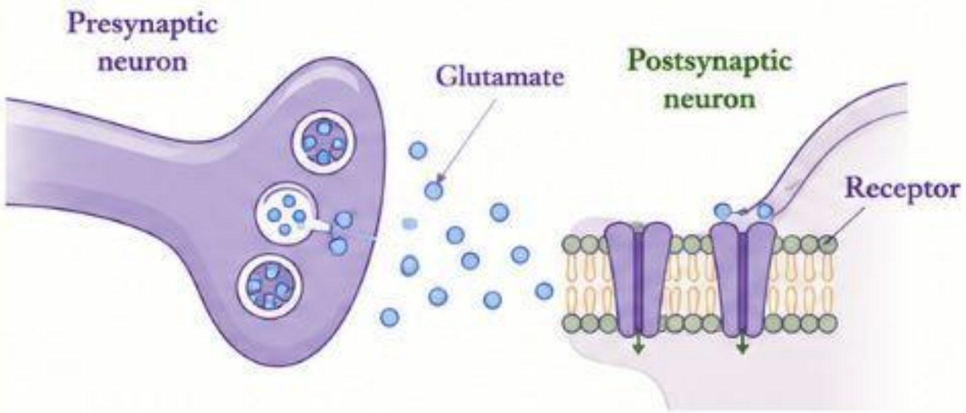
↓

6.

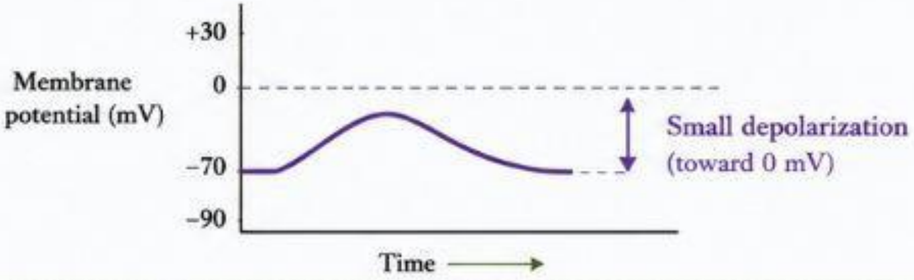
↓

7. **EPSP** An **EPSP** is produced.

What happens at the synapse?



Result: Depolarization



Key Point: An EPSP (Excitatory Postsynaptic Potential) is a small depolarization that makes the neuron **more likely to fire** an action potential.

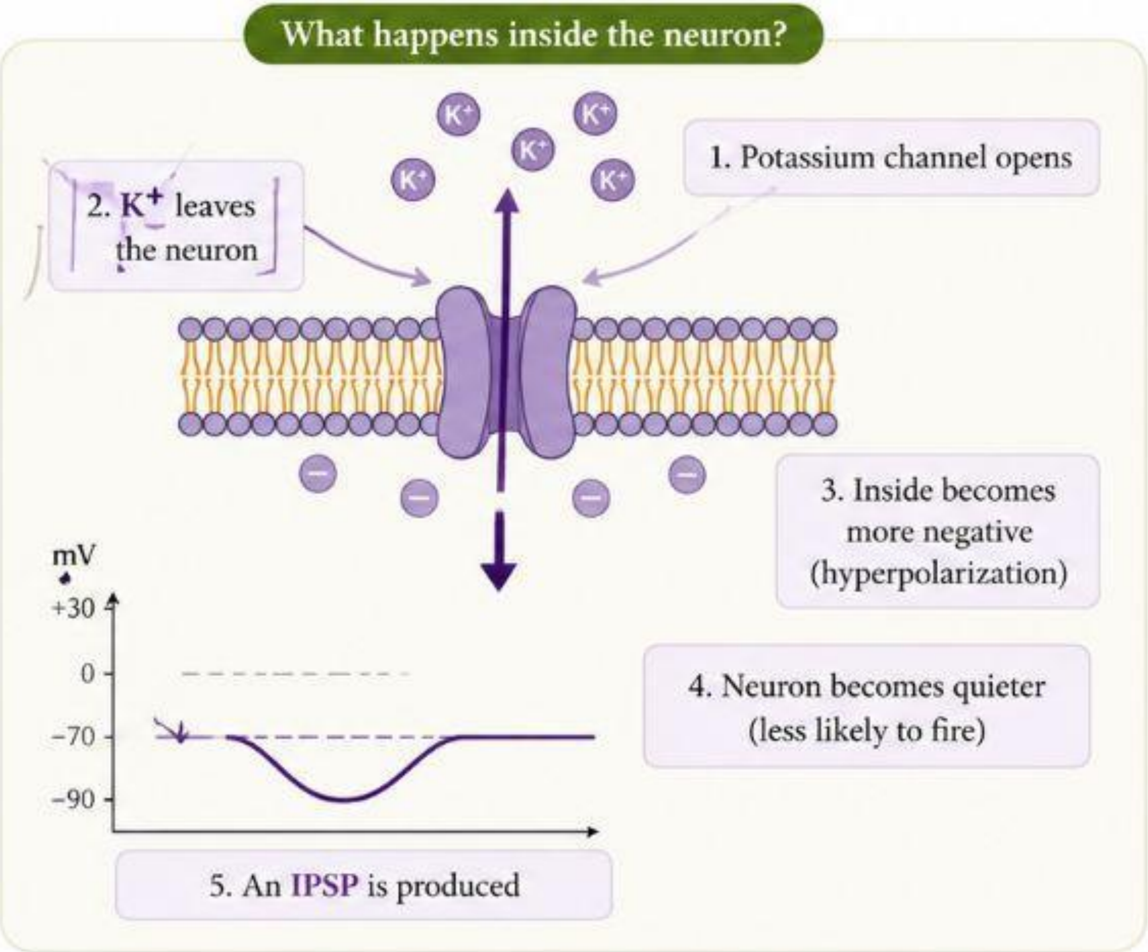
EPSP = **Excitatory Postsynaptic Potential**

IPSP

- An IPSP is a **small electrical change that makes the neuron less likely to fire an action potential.**

How is an IPSP produced?

1. **Potassium channels** open.
2. K^+ leaves the neuron.
3. The inside becomes **more negative.**
4. The neuron becomes **quieter**.
5. An **IPSP** is produced.



Clinical Examples

Excitatory Neurotransmitter

Glutamate

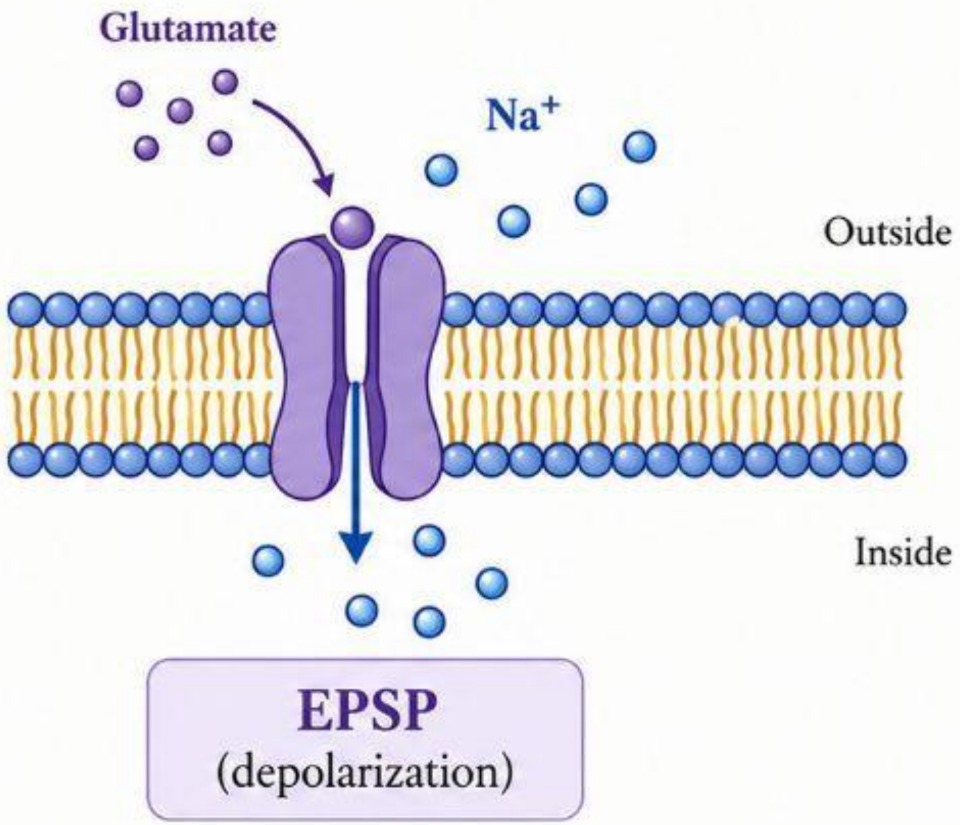
1. Opens Na^+ channels.

↓

2. Na^+ ions enter the neuron.

↓

3. Produces **EPSP**.



Inhibitory Neurotransmitter

GABA

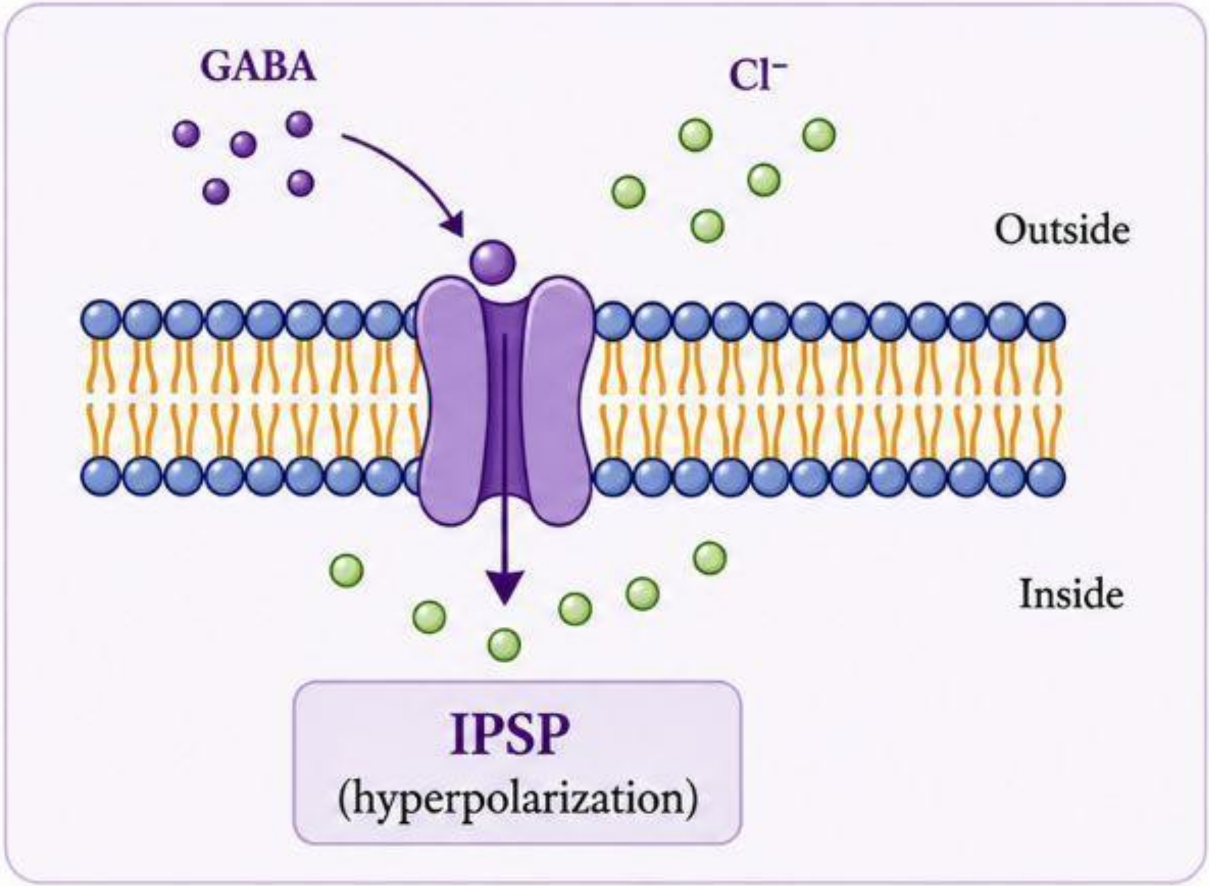
1. Opens Cl^- channels.

↓

2. Cl^- ions enter the neuron.

↓

3. Produces IPSP.



ROLE OF CNS ORGANIZATION

What is CNS Organization?

Organization means **how the neurons in the brain and spinal cord are arranged to perform different functions.**

The CNS is organized into **two major systems**:

Hierarchical System

Diffuse System

Hierarchical System

Organized

Precise

Fast

Controls specific functions

Diffuse System

Wide spread
Less specific

Controls overall brain activity such as sleep, mood, alertness, and attention.

Hierarchical Systems

- It is called **hierarchical** because information moves in an organized pathway from one neuron to another.
- These systems generally contain large myelinated, rapidly conducting fibers.
- Hierarchical systems control major sensory and motor functions.
- The major excitatory transmitters in these systems are aspartate and glutamate.

Some neurotransmitters make neurons **more active**.

These are called **excitatory neurotransmitters**.

The two important excitatory neurotransmitters are: **Glutamate &Aspartate**

They produce **EPSPs**.

These systems also include numerous small inhibitory interneurons which use GABA or glycine as transmitters

Diffuse Systems

- Diffuse or nonspecific systems are broadly distributed, with single cells frequently sending branches to many different areas.
- A widespread network of neurons that influences many areas of the brain at the same time.
- The transmitters in diffuse systems are often amines (norepinephrine, dopamine, serotonin)

TRANSMITTERS AT CENTRAL SYNAPSES

- To be accepted as a neurotransmitter, a chemical must
- (1) be present in higher concentration in the synaptic area than in other areas (ie, must be localized in appropriate areas),
- (2) be released by electrical or chemical stimulation via a calcium-dependent mechanism,

Acetylcholine

Acetylcholine (ACh) was the **first neurotransmitter discovered**.

It is an important chemical messenger that helps neurons communicate with each other.

It is found in:

Brain (CNS)

Peripheral nervous system (PNS)

Neuromuscular junction (skeletal muscles)

- Most CNS responses to ACh are mediated by a large family of G protein-coupled muscarinic M1 receptors that lead to slow excitation when activated.
- Acetylcholine acts on **two types of receptors**. Nicotinic, Muscarinic
- When acetylcholine binds to an **M1 receptor**,

↓

- The neuron becomes **more active (excited).**

Two types of receptors

1. Ionotropic receptor

Acts directly.

Very fast.

Example:

Nicotinic receptor.

2. Metabotropic receptor (G-protein coupled)

Works indirectly.

Uses a **G-protein** inside the cell.

Produces **second messengers**.

Acts more slowly.

The effect lasts longer.

Why is the excitation slow?

Because the receptor does **not open the ion channel directly**.

Instead,

it first activates a **G-protein**.

The G-protein then activates second messengers.

Finally,

the neuron becomes excited.

This takes more time.

Drugs affecting the activity of cholinergic systems in the brain include acetylcholinesterase inhibitors used in Alzheimer's disease (e.g., rivastigmine).

Acetylcholine should not stay in the synapse forever.

It must be removed after it has transmitted the signal.

This is done by an enzyme called:

Acetylcholinesterase (AChE)

What are Acetylcholinesterase Inhibitors?

These drugs **block the enzyme acetylcholinesterase**.

↓

ACh is not broken down.

↓

More ACh remains in the synapse.

↓

More stimulation of M1 receptors.

↓

Better communication between neurons.

Why is Rivastigmine used in Alzheimer's Disease?

In Alzheimer's disease,

many cholinergic neurons die.

↓

Less acetylcholine is produced.

↓

Memory becomes poor.

After giving Rivastigmine

Acetylcholinesterase is blocked.

↓

More acetylcholine remains in the synapse.

↓

Communication between neurons improves.

↓

Memory improves (temporarily).

Dopamine

Dopamine is one of the most important neurotransmitters in the brain.

The major areas producing dopamine are, Substantia nigra, Hypothalamus

Controls Movement, Reward Pathway, Learning and Memory.

Dopamine usually slows down neuron activity.

It does **not work immediately**.

Instead, it works through **G-protein coupled receptors**, so the effect is **slow but long-lasting**.

Drugs that block the activity of dopaminergic pathways include older antipsychotics (eg, chlorpromazine, haloperidol), which may cause parkinsonian symptoms.

Drugs that increase brain dopaminergic activity include CNS stimulants (eg, amphetamine), and commonly used antiparkinsonism drugs (eg, levodopa).

Drug: Pramipexole (Dopamine Agonist)

It acts like dopamine.

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Stimulates dopamine (D2) receptors.

↓

Improves movement.

Use: Parkinson's disease

Drug: Haloperidol

Haloperidol blocks D2 receptors.

↓

Dopamine cannot bind.

↓

Dopamine action decreases.

↓

Used in schizophrenia.

Norepinephrine

Made in:Pons (brainstem), mainly in the **Locus Coeruleus**.

- 🧠 Keeps us **awake and alert**
- 📖 Helps us **focus and learn**
- ❤️ Helps the body respond to **stress (fight or flight)**
- 😊 Helps maintain **normal mood**

α1, β1 → Increase activity (Excitatory)

α2, β2 → Decrease activity (Inhibitory)

Normal Process

Nerve Cell

Releases NE ↓

Synapse ↓

NE binds to receptor ↓

NE is taken back into the nerve cell (Reuptake)

This stops the signal.

Drugs that Increase Norepinephrine

1. Amphetamine

Amphetamine causes more norepinephrine to be released.

Result:

Increased alertness

Increased energy

Increased concentration

Cocaine

Normally, norepinephrine is taken back into the nerve cell after it is released.

Cocaine blocks this reuptake.

As a result:

More norepinephrine stays in the synapse.

↓

Stronger stimulation.

Serotonin

- **Produced in:** Raphe nuclei of the **pons and upper brainstem**.
- **Functions:**
  - 😊 Regulates **mood**
  - 😴 Controls **sleep**
  - 🍽️ Regulates **appetite**
  - 😌 Maintains **emotional balance**
- **Low serotonin** → Depression, anxiety, poor sleep.

SSRIs (Selective Serotonin Reuptake Inhibitors)

Examples:

Fluoxetine

Sertraline

Escitalopram

Paroxetine

How do SSRIs work?

Normally, after serotonin is released,it is taken back into the nerve cell (reuptake).

SSRIs **block this reuptake**.

↓

More serotonin stays in the synapse.

↓

Mood improves.

Glutamic Acid

- **Glutamic acid (Glutamate)** is the **major excitatory neurotransmitter** in the brain. Most neurons in the brain are excited by glutamic acid.
- Glutamate is important for: Learning, Memory, Thinking
- There are several glutamate receptors. include the N-methyl-d-aspartate (NMDA)
- When glutamate binds to NMDA receptors, Calcium enters the neuron, The neuron becomes stronger, Learning and memory improve.

Drugs that Block NMDA Receptor

PCP (Phencyclidine)

Ketamine

These drugs block NMDA receptors.

↓

Glutamate cannot work properly.

↓

Learning and memory decrease.

Memantine

Memantine is an NMDA antagonist used in Alzheimer's dementia.

Memantine partially blocks NMDA receptors.

Why?

Because in Alzheimer's disease,

too much glutamate can damage brain cells.

Memantine protects neurons from this damage.

GABA and Glycine

- **GABA (Gamma Aminobutyric Acid)** is the **major inhibitory neurotransmitter** in the brain.
- It prevents excessive firing of neurons.

GABA released

↓

Binds GABA receptor

↓

Chloride enters neuron

↓

Neuron becomes less active

↓

No action potential

Why is GABA Important?

Without GABA, neurons would fire continuously.

This may cause:

Anxiety

Seizures

Muscle spasms

Tremors

Clinical Importance
Anxiety Disorders

How do Benzodiazepines work?

They increase the action of GABA.

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Brain activity decreases.

↓

Patient becomes relaxed.

↓

Anxiety decreases.